ISE492 - Graduation Project 2

AI Based Navigation Recommendation System

ShouldAI

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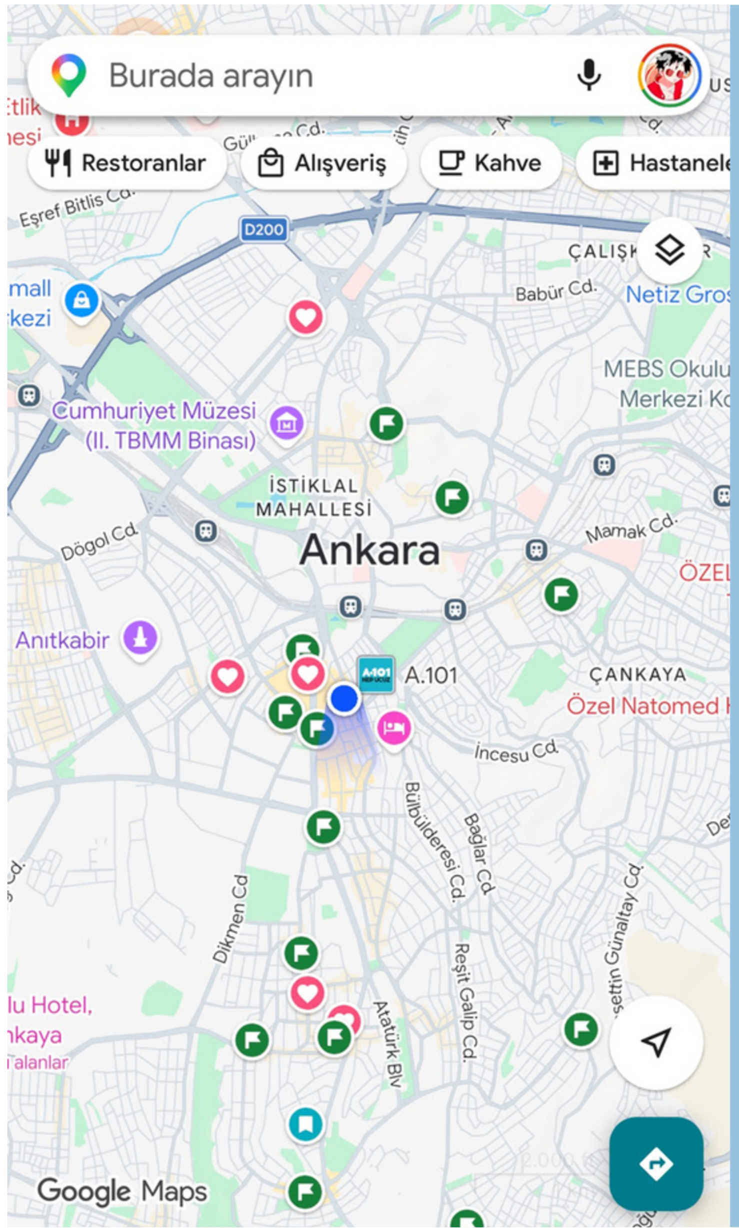
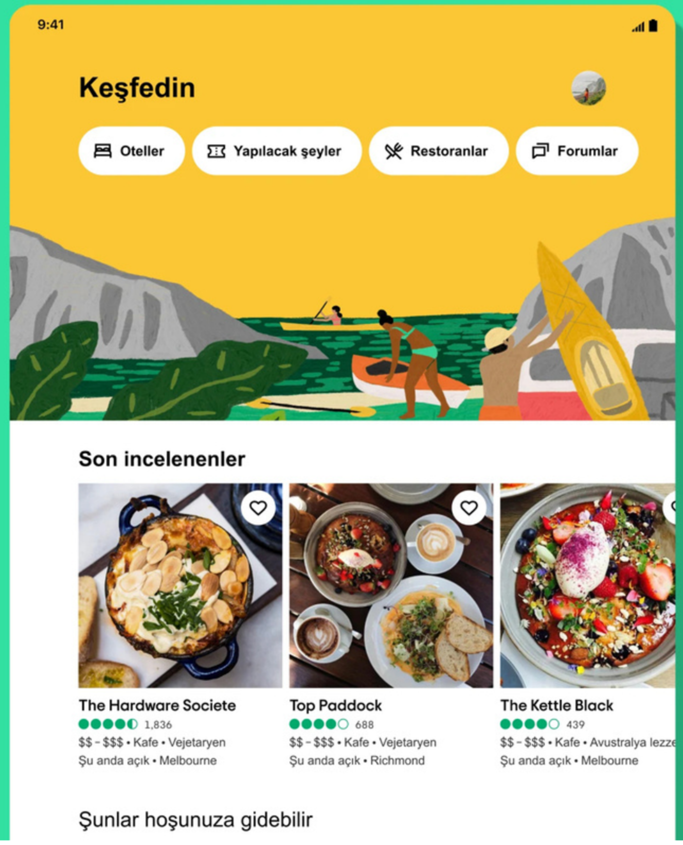
Literature Review



Navigations like Google Maps, Yandex Navi, Apple Maps...

Strong routing and huge POI data, but passive; it only reacts when the user searches. Basically manual choices and distraction during travels.

Bir sonraki seyahatinizi keşfedin ve planlayın



Rating websites like Tripadvisor, Foursquare, Tasteatlas...

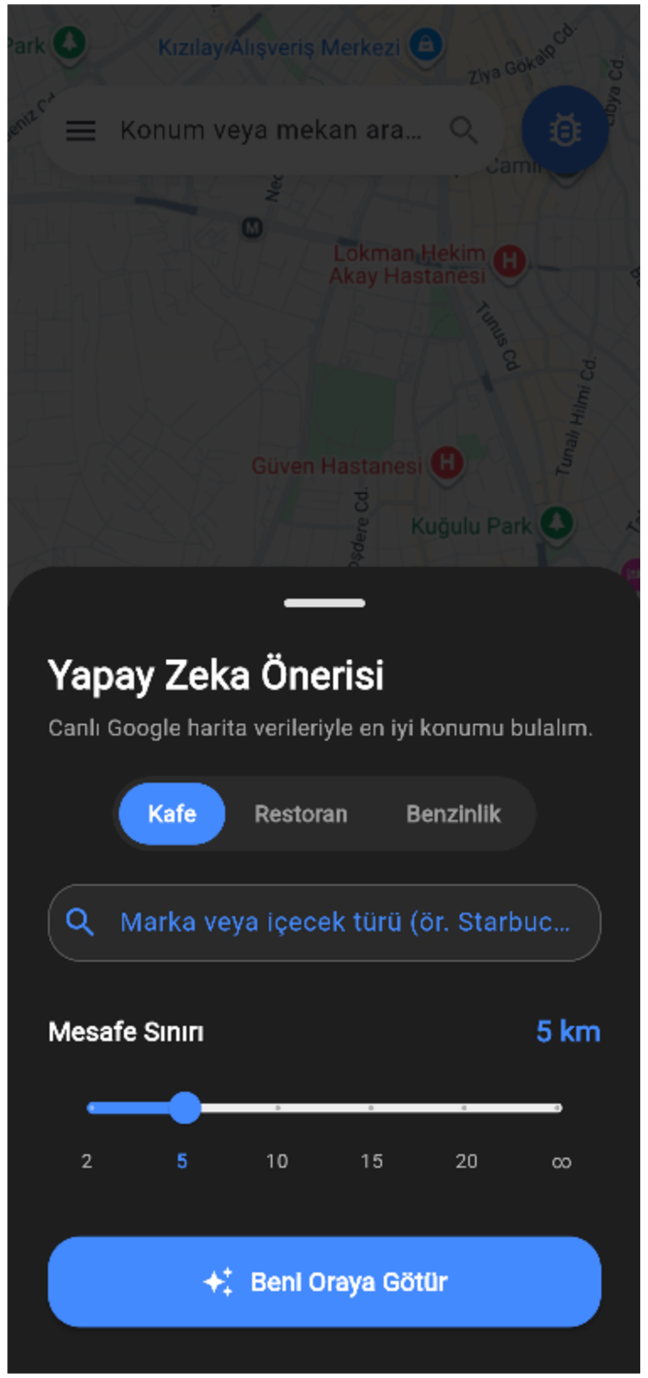
Rich reviews and ratings, but disconnected from driving; the user must switch apps and re-enter the place manually. Seperated service.

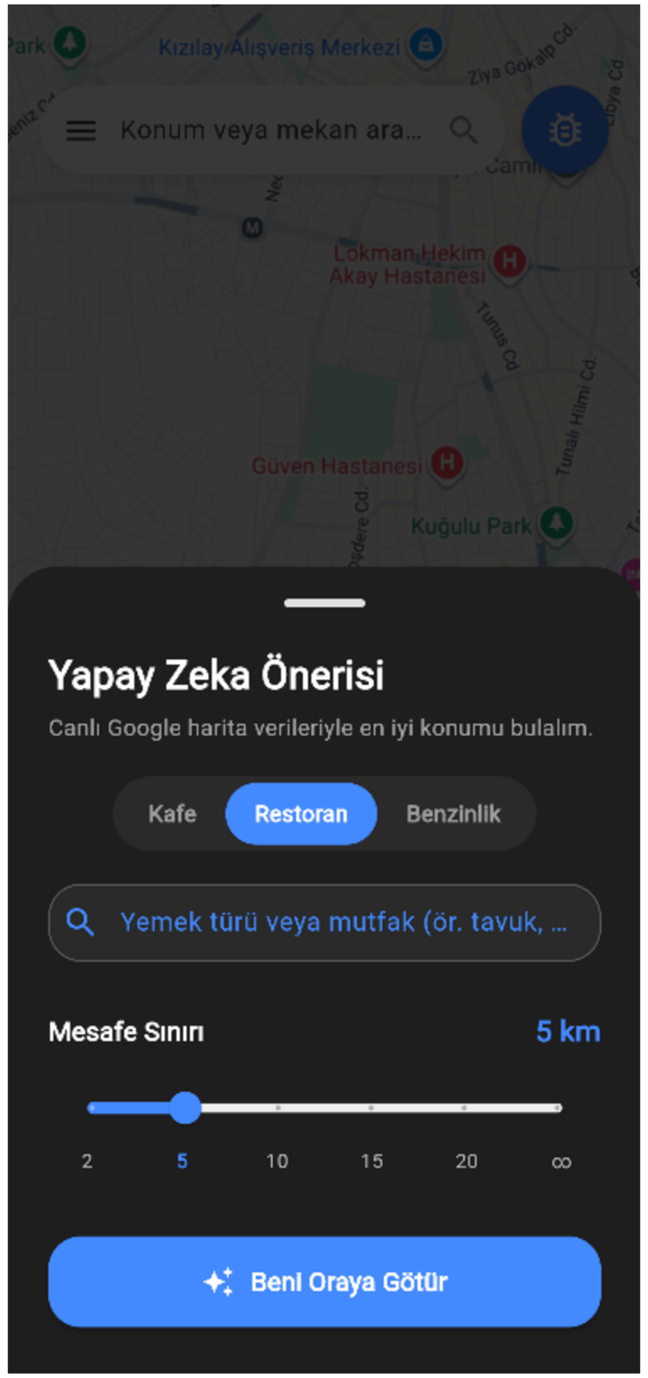
The Gap:

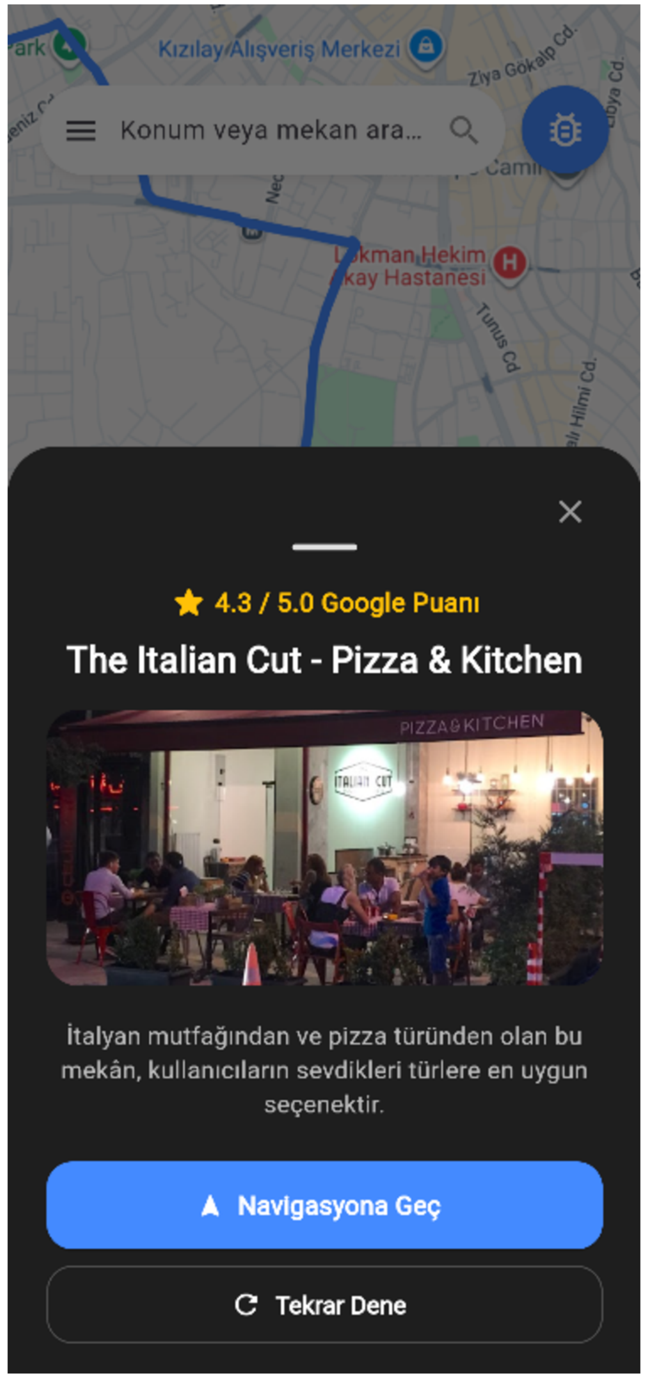
No tool routes and recommends proactively and personally during the trip.

Innovations We Did

We put both services together to avoid unnecessary time and effort.
Now users can reach the ideal places for their needs.

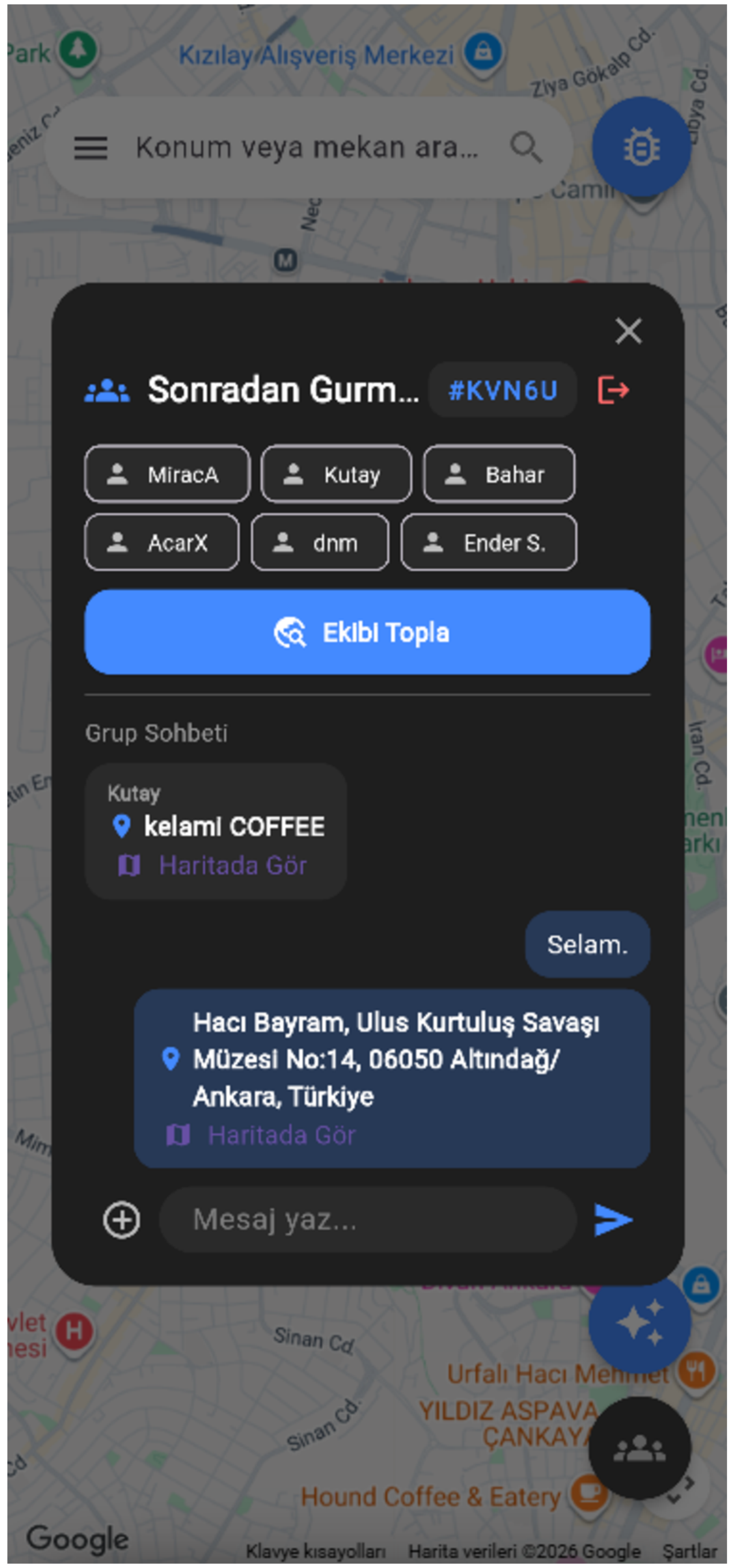






NEW Group System and Features

- 5-character group code
- Real-time Chat
- Location & Place Sharing
- Gather the Team



System Architecture

Frontend — Flutter (web): UI, live location, map → Netlify
Why: single codebase for any platform

Backend — FastAPI (Python): recomm. engine → Render
Why: full control of the logic · native Python AI ecosystem

Database — Supabase's PostgreSQL, Auth, Realtime
Why: relational fit for groups & preferences · built-in auth + realtime · efficient free tier

External — Google Maps & Places + Groq LLM
Why: best location data in Türkiye · fast, low-cost LLM inference

How It Works

Retrieve — query Google Places by category or dish within a dynamic radius

Filter — open anytime, 50+ reviews, category-type, route-forward

Personalize — liked/disliked cuisines, green/gray matching, Smart & Selective modes

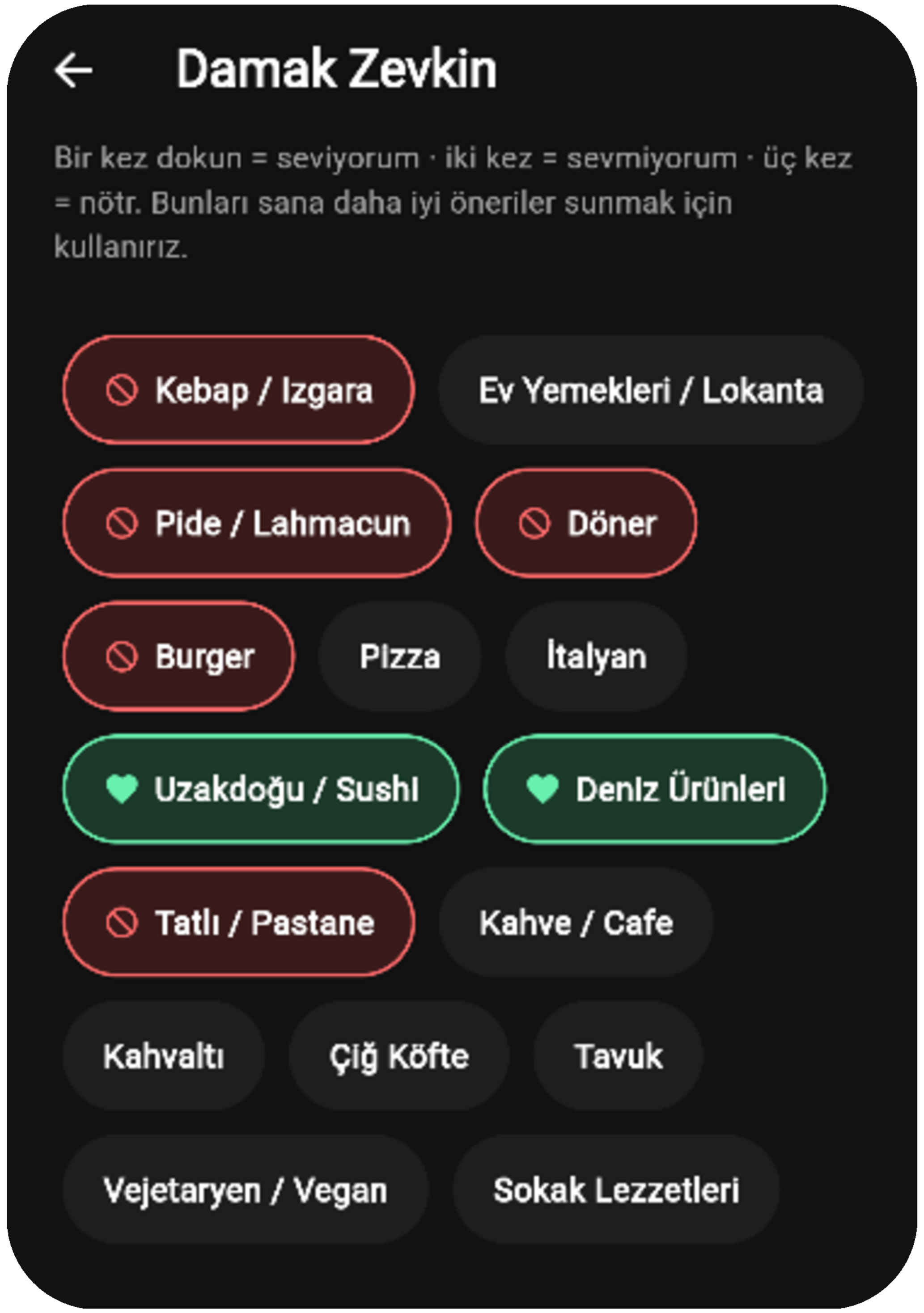
AI Decision — Groq LLM picks the best match for the user's intent and explains it

Present — non-intrusive card, one-tap waypoint, no distraction during ride

Customized Pop-ups

Green: Cuisines the user likes — the system actively prefers and recommends these.

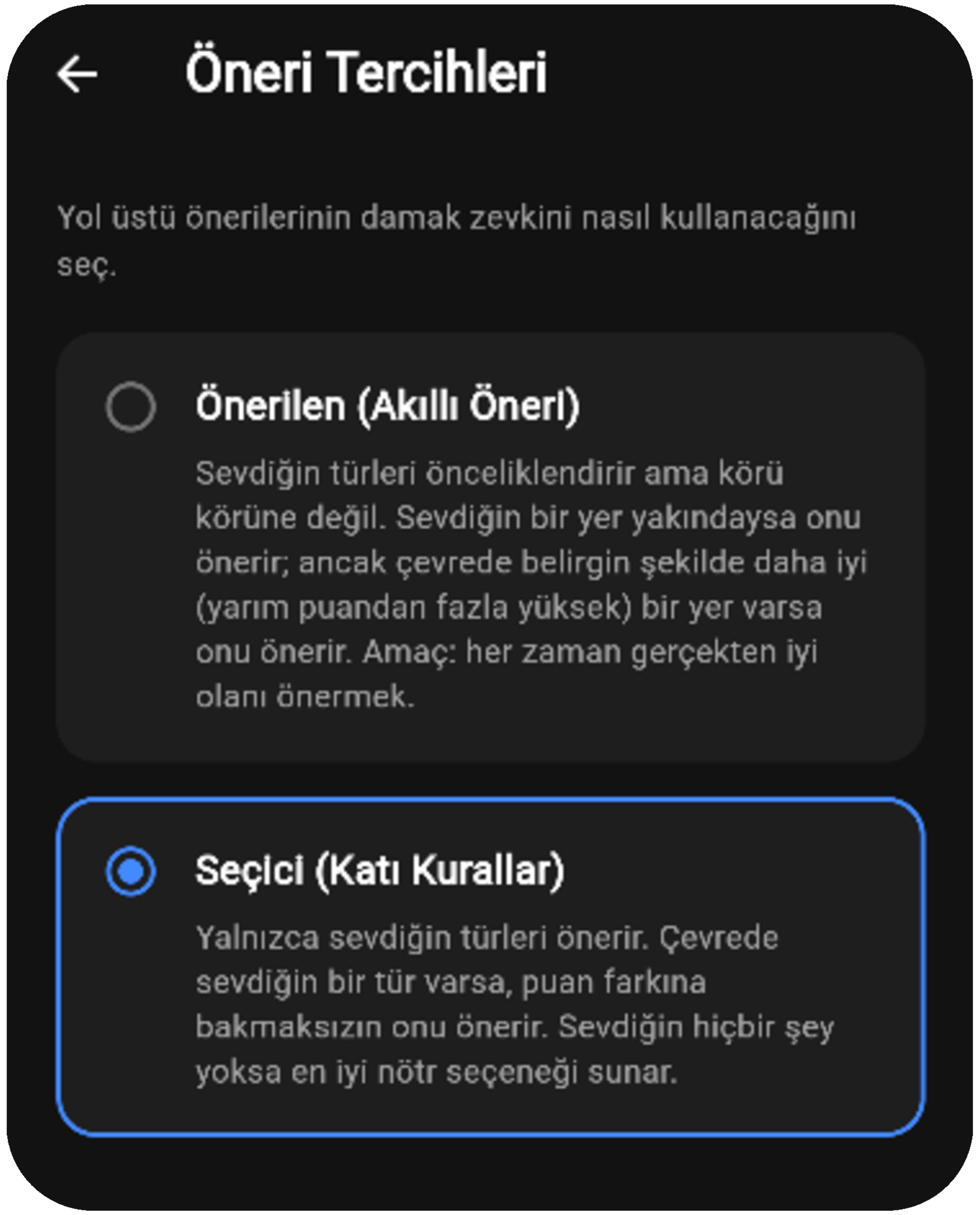
Red: Cuisines the user dislikes — the system filters these out so they're never recommended.



Recommendation Algorithms

Default (Smart) - Prefers liked cuisines, but switches to a neutral place if it's rated 0.5+ higher nearby.

Selective — Recommends only liked cuisines; if none nearby, falls back to the best place with a note.



LIVE DEMO

Let’s test what we talked.

Thanks for your attention.